

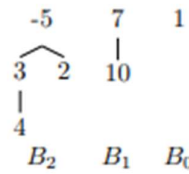
Discipline: COMPUTER SCIENCE AND ENGINEERING**Course Code & Name: 222TCS100 ADVANCED DATA STRUCTURES AND ALGORITHMS****Max. Marks: 60****Duration: 2.5 Hours****PART A*****Answer all questions. Each question carries 5 marks*****Marks**

- | | | |
|---|--|-----|
| 1 | Illustrate graphically the various asymptotic notations with example. | (5) |
| 2 | Explain how disjoint set data structure is used to find connected components on an undirected graph.. | (5) |
| 3 | Consider the bipartite graph $G(L \cup R, E)$ where $L = \{1, 2, 3, 4, 5\}$ and $R = \{A, B, C, D, E\}$ with edges $E = \{(1, A), (1, B), (2, B), (2, C), (3, D), (4, E), (5, E)\}$. Find the maximum bipartite matching of this graph. | (5) |
| 4 | Apply Miller Rabin algorithm to test whether the number 341 is prime or not. | (5) |
| 5 | Demonstrate the approximation algorithm for the subset-sum problem. | (5) |

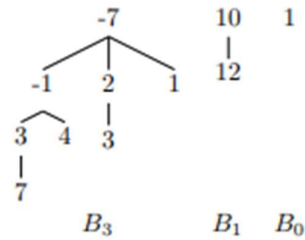
PART B***Answer any 5 questions. Each question carries 7 marks***

- | | | |
|---|---|------------|
| 6 | (a) Given a text $T = \text{"abcxabc dabc dabc y"}$ and a pattern $P = \text{"abc dabc y"}$, use the Knuth-Morris-Pratt algorithm to find all occurrences of P within T .
(b) Compute the amortized cost of incrementing a binary counter using potential method. | (4)
(3) |
| 7 | (a) Apply the Rabin-Karp algorithm to search for the pattern "ABC" in the text "ABCDABCDEABC" using a prime base $q = 101$.
(b) Explain aggregate method of amortized analysis with a suitable example. | (4)
(3) |
| 8 | (a) Analyse the time complexity of decrease-key operation in Fibonacci heap.
(b) Illustrate merge operation on the binomial heaps H_1 and H_2 shown in the figure. | (3)
(4) |

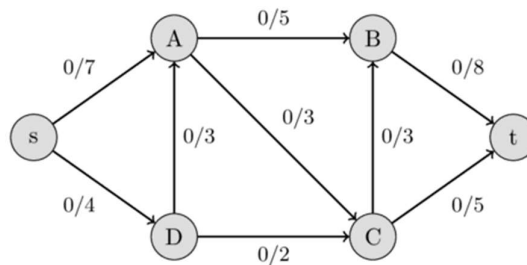
Binomial heap H_1



Binomial heap H_2



- 9 Describe Ford Fulkerson algorithm and find the maximum flow for the flow network shown below (edge labels are of the form flow/capacity). (7)



- 10 (a) State and prove Fermat's theorem. (3)
 (b) Analyse randomized quicksort and derive its running time. (4)
- 11 Verify the correctness of the matrix multiplication, $AB = C$ where, (7)

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \quad B = \begin{pmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 30 & 24 & 18 \\ 84 & 69 & 54 \\ 138 & 114 & 90 \end{pmatrix}$$

- 12 (a) Explain the approximation algorithm for traveling salesperson problem. (4)
 (b) Compare the optimal vertex cover and the approximate vertex cover produced by Greedy based approximation algorithm, for the undirected graph $G = (V, E)$ shown below: (3)

